



LUCAS NOIROT (PETIT)

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🌐 enpeluche 📧 enpeluche 📄 enpeluche.github.io

📍 Montpellier, France

🚗 Permis B

PROFILE

Passionate about Computer Science and Mathematics, I specialize in bridging the gap between abstract concepts and concrete, practical solutions.

Personal Attributes

Rigorous

Abstract Thinking

Self-learner

Proactive

Team Player

Creative

Interests

Origami

Board Games

Astronomy

TECHNICAL SKILLS

Formal Methods & Cryptography

Formal Methods (Coq)

Security Proofs

Optimization & Modeling

Mixed-Integer Programming (MIP)

Constraint Programming

Heuristics

Complexity Analysis

Development

C/C++

Rust

OCaml

Python

SageMath

Java

HTML/CSS/JavaScript

Tools & Environment

Linux

Git

Docker

LaTeX

LANGUAGES

- French (Native)
- English (C1 – Full Professional Proficiency)

R&D ENGINEER - ALGORITHMS & APPLIED MATHEMATICS

EDUCATION

📅 2025 M.SC. IN COMPUTER SCIENCE – ALGORITHMS

📍 University of Montpellier, France

Cryptography – Computer Algebra – Optimization

📅 2023 M.SC. IN MATHEMATICS (1ST YEAR)

📍 University of Burgundy, France

Linear Algebra – Field Theory – Probability Theory

📅 2022 B.SC. IN MATHEMATICS

📍 University of Burgundy, France

2019 Ranked 5th out of 55 (Top 10%)

EXPERIENCE

📅 2025 RESEARCH INTERNSHIP IN CRYPTOGRAPHY

(5 months) 📍 LIRMM Laboratory – ECO Team

Topic: Lattice Reduction – Adapting strategies from the polynomial case. Developed reduction algorithms using **SageMath**. Conducted complexity analysis and benchmarked performance between polynomial and integer arithmetic.

🔗 enpeluche.github.io/lattice-reduction-internship/

📅 2022 PRIVATE TUTOR

📍 Dijon, France – Freelance

Provided Mathematics and Physics tutoring to middle and high school students.

PERSONAL PROJECTS

Workforce Scheduling & Routing (WSRP)

Designed a hybrid solver for technician routing under skill constraints. Implemented a **heuristic** in **Java** coupled with Constraint Programming (**OR-Tools**).

🔗 [enpeluche/wsrp-lns-solver](https://enpeluche.github.io/wsrp-lns-solver)

Coq-Verified Red-Black Tree

Formalized and proved balancing invariants in **Coq**, with automatic extraction to an efficient, certified **OCaml** module.

🔗 [enpeluche/coq-verified-rbt](https://enpeluche.github.io/coq-verified-rbt)

Constraint Solving Engine Architecture (CSP)

Developed a propagation engine from scratch in **C++**. Implemented Arc-Consistency algorithms (**AC-3**).

🔗 [enpeluche/constraint-puzzle-solver](https://enpeluche.github.io/constraint-puzzle-solver)

Zero-Knowledge Proof (ZKP) Protocol

Implemented the **Schnorr authentication protocol** over Elliptic Curves. Prototyped algebraically in **SageMath** and developed a high-performance version in **C++**.

🔗 [enpeluche/zkp-schnorr-cpp](https://enpeluche.github.io/zkp-schnorr-cpp)

Post-Quantum Messaging (Lattice-based)

Implemented a quantum-resistant encrypted chat application. Developed a **C++** key exchange protocol based on the **LWE (Learning With Errors)** problem over lattices. Built an asynchronous Client/Server architecture.

 <https://github.com/enpeluche/>